Sampling From Two Independent Normal populations.

Let X and Y be two independently distributed r.v.s with distribution $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ respectively. Let $x_1, x_2, \ldots x_m$ be a random sample of size m from X and Let $y_1, y_2 \ldots y_n$ be another random sample of size n from Y. We wish to test the equality of Means and equality of Variances.

1. Testing of Equality of Means:

- Assuming $\sigma_1^2 = \sigma_2^2 = \sigma^2$ in the two populations. We wish to test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$. Now,

$$f(x_i, \mu_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i - \mu_1)^2}$$

$$f(y, \mu_2, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - \mu_2)^2}$$

the

$$L(\underline{x}, \mu_1, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{m/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i - \mu_1)^2}$$

$$L(\underline{y}, \mu_2, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu_2)^2}$$

since X & Y are independent,

$$f(x, y, \mu_1, \mu_2, \sigma^2) = f(x, \mu_1, \sigma^2) f(y, \mu_2, \sigma^2)$$

and

$$L(\underline{x}, \underline{y}, \mu_1, \mu_2, \sigma^2) = \left(\frac{1}{2\pi}\right)^{\frac{m+n}{2}} \left(\frac{1}{\sigma^2}\right)^{m+n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i - \mu_1)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu_2)^2}$$

Now,

$\Omega = \{(\mu_1, \mu_2, \sigma^2); -\infty < \mu_1, \mu_2 < \infty, \sigma^2 > 0\}$

$\Omega_0 = \{(\mu_1, \mu_2, \sigma^2); \mu_1 = \mu_2 = \mu(\text{say}), \sigma^2 > 0\}$

Therefore the Maximum likelihood estimate of μ_1, μ_2, σ^2 are:

$$\hat{\mu}_1 = \bar{x}, \quad \hat{\mu}_2 = \bar{y}, \quad \hat{\sigma}^2 = \frac{1}{m+n}\left[\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2\right]$$

$$\Rightarrow L(\hat{\Omega}) = \left(\frac{1}{2\pi}\right)^{\frac{m+n}{2}}\left(\frac{1}{\hat{\sigma}^2}\right)^{\frac{m+n}{2}} e^{-\frac{1}{2\hat{\sigma}^2}\left[\sum_{i=1}^{m}(x_i-\hat{\mu}_1)^2 + \sum_{i=1}^{n}(y_i-\hat{\mu}_2)^2\right]}$$

$$= \left(\frac{1}{2\pi}\right)^{m+n/2}\left[\frac{m+n}{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{m+n/2} e^{-\frac{1}{2}\frac{\left[\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2\right]}{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

$$= \left(\frac{1}{2\pi}\right)^{m+n/2}\left[\frac{m+n}{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{m+n/2} e^{-(m+n)/2}$$

Now in

Ω_0, $\mu_1 = \mu_2 = \mu$ (say) then H_0 is true.

The

$$L = \left(\frac{1}{2\pi}\right)^{m+n/2}\left(\frac{1}{\sigma^2}\right)^{\frac{m+n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i-\mu)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2}$$

Then the M.L estimates of μ, σ^2 are:

$$\hat{\mu} = \frac{\sum x_i + \sum y_i}{m+n} = \frac{m\bar{x}+n\bar{y}}{m+n}$$

and

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{m}(x_i-\hat{\mu})^2 + \sum_{i=1}^{n}(y_i-\hat{\mu})^2}{m+n}$$

$$= \frac{\sum_{i=1}^{m}\left(x_i - \frac{m\bar{x}+n\bar{y}}{m+n}\right)^2 + \sum_{i=1}^{n}\left(y_i - \frac{m\bar{x}+n\bar{y}}{m+n}\right)^2}{m+n}$$

Note:

$$\sum_{i=1}^{m}\left(x_i - \left(\frac{m\bar{x}+n\bar{y}}{m+n}\right)\right)^2 = \sum_{i=1}^{m}\left[\underbrace{(x_i-\bar{x})}_{a} + \left(\bar{x} - \frac{m\bar{x}+n\bar{y}}{m+n}\right)\right]^2$$

$$= \sum_{i=1}^{m}(x_i-\bar{x})^2 + m\left(\bar{x} - \frac{m\bar{x}+n\bar{y}}{m+n}\right)^2 = \sum_{i=1}^{m}(x_i-\bar{x})^2 + \frac{mn^2}{(m+n)^2}(\bar{x}-\bar{y})^2$$

Similarly, $\sum_{i=1}^{n}\left(y_i - \left(\frac{m\bar{x}+n\bar{y}}{m+n}\right)\right)^2 = \sum_{i=1}^{n}(y_i-\bar{y})^2 + \frac{m^2n}{(m+n)^2}(\bar{x}-\bar{y})^2$

Hence:

$$= \frac{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2 + \frac{mn}{m+n}(\bar{x}-\bar{y})^2}{m+n}$$

$$\Rightarrow L(\hat{\omega}_0) = \left(\frac{1}{2\pi}\right)^{m+n/2} \left[\frac{m+n}{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2 + \frac{mn}{m+n}(\bar{x}-\bar{y})^2}\right]^{m+n/2} * e^{-(m+n)}$$

$$\Rightarrow \lambda = \frac{L(\hat{\omega}_0)}{L(\hat{\omega})} = \left[\frac{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2 + \frac{mn}{m+n}(\bar{x}-\bar{y})^2}\right]^{\frac{m+n}{2}} * e^{-(m+n)/2}$$

$$\lambda = \left[\frac{1}{1 + \frac{mn}{m+n}(\bar{x}-\bar{y})^2 \Big/ \sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{\frac{m+n}{2}}$$

Now $\bar{x}-\bar{y}$ is normally distributed with Mean

$E(\bar{x}-\bar{y}) = \mu_1-\mu_2$ and $Var(\bar{x}-\bar{y}) = Var(\bar{x}) + Var(\bar{y}) - 2Cov(\bar{x},\bar{y})$

$= Var(\bar{x}) + Var(\bar{y})$ since $Cov(\bar{x},\bar{y}) = 0$

$= \frac{\sigma^2}{m} + \frac{\sigma^2}{n}$

$\therefore \bar{x}-\bar{y} \sim N\left(\mu_1-\mu_2, \sigma^2\left(1/m + 1/n\right)\right)$

$$\Rightarrow Z = \frac{(\bar{x}-\bar{y})-(\mu_1-\mu_2)}{\sqrt{\sigma^2(1/m+1/n)}} = \frac{(\bar{x}-\bar{y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma^2(m+n)}{mn}}}$$

$$= \sqrt{\frac{mn}{m+n}}\left[\frac{(\bar{x}-\bar{y})-(\mu_1-\mu_2)}{\sigma}\right] \sim N(0,1)$$

also $\frac{\sum_{i=1}^{m}(x_i-\bar{x})^2}{\sigma^2} \sim \chi^2_{m-1}$ and $\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{\sigma^2} \sim \chi^2_{n-1}$

$$\Rightarrow U = \frac{\sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2}{\sigma^2} \sim \chi^2_{m+n-2}$$

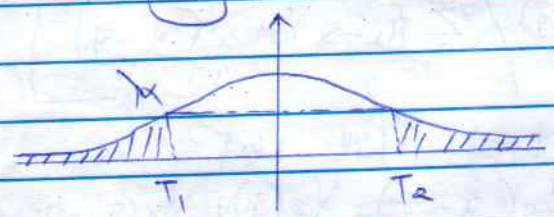
Therefore when H_0 is true the statistic

$$T = \frac{\sqrt{\frac{mn}{m+n}}\,(\bar{x}-\bar{y})}{\sqrt{\frac{\sum_{i=1}^{m}(x_i-\bar{x})^2+\sum_{i=1}^{n}(y_i-\bar{y})^2}{\sigma^2(m+n-2)}}} = \sim t_{m+n-2}$$

Therefore

$$\lambda = \left[\frac{1}{1+T^2/(m+n-2)}\right]^{m+n/2}$$

Which is a decreasing function of T^2.
plotting λ against T we've,



The shaded region is the critical region. Hence we reject H_0 whenever $T < T_1$ or $T > T_2$ where T_1 and T_2 are constants satisfying,

$$P[T < T_1 \text{ or } T > T_2 / H_0] = \alpha$$

for an equal tailed T_1 and T_2 satisfy

$$P[T < T_1 / H_0] = \alpha/2 = P[T > T_2 / H_0]$$

$$T_1 = -t_{\alpha/2} \text{ and } T_2 \geq t_{\alpha/2}$$

where $t_{\alpha/2}$ satisfies $P[t_{m+n-2} > t_{\alpha/2}] = \alpha/2$

Therefore the LR test for $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$ is to compute T and reject H_0 at α level of significance whenever

$$T < -t_{\alpha/2} \text{ or } T > t_{\alpha/2} \text{ or simply } |T| \geq t_{\alpha/2}.$$

Example:

A botanist is interested in comparing the growth responses for dwarf pea stem to two different levels of hormone indoleacetic acid. Let X denote the growth attributed to level I on the stem and Y to level II. 11 Observations on X were taken and 13 in Y as follows:

X: 0.8, 1.0, 0.1, 0.9, 1.7, 1.0, 1.4, 0.9, 1.2, 0.5, 1.8

Y: 1.0, 0.8, 1.6, 2.6, 1.3, 1.1, 2.4, 1.8, 2.5, 1.4, 1.9, 2.0, 1.2

Assuming there are random samples from normal population where $X \sim N(\mu_1, \sigma^2)$ and and $Y \sim N(\mu_2, \sigma^2)$ i.e $\sigma_1^2 = \sigma_2^2 = \sigma^2$, test the hypotheses

$H_0: \mu_1 = \mu_2 = \mu$ against

$H_1: \mu_1 \neq \mu_2$, take $\alpha = 0.05$

Solution:

The test statistic $T = \dfrac{\sqrt{\frac{mn}{m+n}}\left(\bar{x} - \bar{y}\right)}{\sqrt{\dfrac{\sum_{i=1}^{m}(x_i - \bar{x})^2 + \sum_{i=1}^{n}(y_i - \bar{y})^2}{m+n-2}}}$

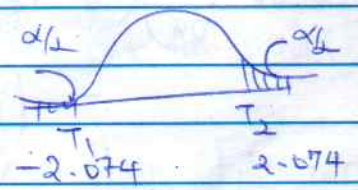
From the data above, we obtain the following statistics

$\bar{x} = 1.027$, $\sum_{i=1}^{m}(x_i - \bar{x})^2 = 2.442$, $m = 11$

$\bar{y} = 1.662$, $\sum_{i=1}^{n}(y_i - \bar{y})^2 = 4.231$, $n = 13$

Hence $T = \dfrac{\sqrt{\frac{11 \times 13}{11+13}}\left(1.027 - 1.662\right)}{\sqrt{\dfrac{2.442 + 4.231}{11+13-2}}} = -2.81$

from the table $t_{22, 0.025} = 2.074$



$|T| = |-2.81| > 2.074$ Thus we reject H_0 at 5% level of significance.

Tests on the Variances of two normal populations:

Let X and Y be two independent r.vs, where $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$

Suppose we want to test the hypothesis $H_0: \sigma_1^2 = \sigma_2^2 = \sigma^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$

Let $x_1, x_2 \ldots x_m$ and $y_1, y_2 \ldots y_n$ be two independent random samples from X and Y respectively. The likelihood function is:

$$L(\Omega) = \left(\frac{1}{2\pi\sigma_1^2}\right)^{m/2} \left(\frac{1}{2\pi\sigma_2^2}\right)^{n/2} e^{-\frac{1}{2\sigma_1^2}\sum_{i=1}^{m}(x_i-\mu_1)^2 - \frac{1}{2\sigma_2^2}\sum_{i=1}^{n}(y_i-\mu_2)^2}$$

The parameter spaces are:

$$\Omega = \{(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2): -\infty < \mu_1, \mu_2 < \infty,\ \sigma_1^2, \sigma_2^2 > 0\}$$

and $\Omega_0 = \{(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2): -\infty < \mu_1, \mu_2 < \infty,\ \sigma_1^2 = \sigma_2^2 = \sigma > 0\}$

The M.L. estimates of $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ under Ω are

$$\hat{\mu}_1 = \bar{x},\quad \hat{\mu}_2 = \bar{y},\quad \hat{\sigma}_1^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i-\bar{x})^2,\quad \hat{\sigma}_2^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i-\bar{y})^2$$

Hence

$$L(\hat{\Omega}) = \left(\frac{1}{2\pi}\right)^{m+n/2} \left(\frac{1}{\hat{\sigma}_1^2}\right)^{m/2} \left(\frac{1}{\hat{\sigma}_2^2}\right)^{n/2} e^{-m+n/2}$$

Under the Ω_0, the M.L. estimates of $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ are:

$$\hat{\mu}_1 = \bar{x},\quad \hat{\mu}_2 = \bar{y}\quad \hat{\sigma}^2 = \sum_{i=1}^{m}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(y_i-\bar{y})^2 / m+n$$

Hence

$$L(\hat{\Omega}_0) = \left(\frac{1}{2\pi}\right)^{m+n/2} \left(\frac{1}{\hat{\sigma}^2}\right)^{m+n/2} e^{-\frac{1}{2\hat{\sigma}^2}\sum_{i=1}^{m}(x_i-\bar{x})^2 - \frac{1}{2\hat{\sigma}^2}\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \left(\frac{1}{2\pi}\right)^{m+n/2} \left(\frac{1}{\hat{\sigma}^2}\right)^{m+n/2} e^{-(m+n)/2}$$

The likelihood ratio is:

$$\lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})} = \frac{(\hat{\sigma}_1^2)^{m/2}(\hat{\sigma}_2^2)^{n/2}}{(\hat{\sigma}^2)^{m+n/2}}$$

$$= \left[\sum_{i=1}^{m}(x_i-\bar{x})^2\right]^{m/2}\left[\sum_{i=1}^{n}(y_i-\bar{y})^2\right]^{n/2}(m+n)$$

$$m^{m/2}\, n^{n/2}\left[\sum_{i=1}^{m}(x_i-\bar{x})^2+\sum_{i=1}^{n}(y_i-\bar{y})^2\right]^{m+n/2}$$

$$\lambda = \frac{(m+n)^{m+n/2}}{m^{m/2}\, n^{n/2}}\left[\frac{\sum_{i=1}^{m}(x_i-\bar{x})^2}{\sum_{i=1}^{m}(x_i-\bar{x})^2+\sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{m/2}\left[\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{\sum_{i=1}^{m}(x_i-\bar{x})^2+\sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{n/2}$$

$$= \frac{(m+n)^{m+n/2}}{m^{m/2}\, n^{n/2}}\left[\frac{\sum_{i=1}^{m}(x_i-\bar{x})^2/\sum_{i=1}^{n}(y_i-\bar{y})^2}{1+\sum_{i=1}^{m}(x_i-\bar{x})^2/\sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{m/2}\left[\frac{1}{1+\sum_{i=1}^{m}(x_i-\bar{x})^2/\sum_{i=1}^{n}(y_i-\bar{y})^2}\right]^{n/2}$$

Now let $F = \dfrac{\sum_{i=1}^{m}(x_i-\bar{x})^2/(m-1)}{\sum_{i=1}^{n}(y_i-\bar{y})^2/(n-1)}$

The above Variable gives:

$$\lambda = \frac{(m+n)^{m+n/2}}{m^{m/2}\, n^{n/2}}\left[\frac{\frac{m-1}{n-1}F}{1+\frac{m-1}{n-1}F}\right]^{m/2}\left[\frac{1}{1+\frac{m-1}{n-1}F}\right]^{n/2}$$

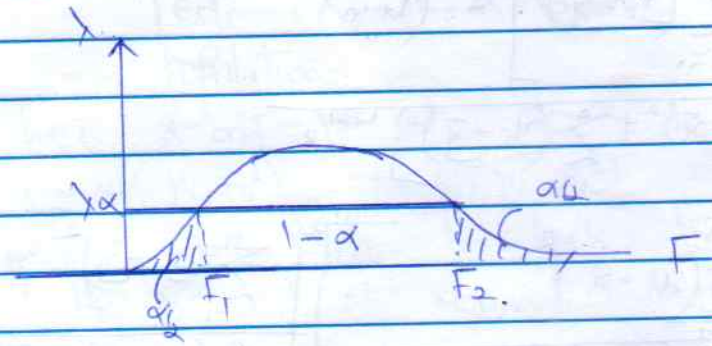
$$= \frac{(m+n)^{m+n/2}}{m^{m/2}\, n^{n/2}}\,\frac{\left(\frac{m-1}{n-1}F\right)^{m/2}}{\left(1+\frac{m-1}{n-1}F\right)^{m+n/2}}$$

The Variable $V = \dfrac{\sum(x_i-\bar{x})^2}{\sigma_1^2(m-1)} \Big/ \dfrac{\sum(y_i-\bar{y})^2}{\sigma_2^2(n-1)}$ has a Fisher's F distribution with $(m-1)$ and $(n-1)$ d.f

Now under H_0:

$$F = \frac{\sum(x_i-\bar{x})^2/(m-1)}{\sum(y_i-\bar{y})^2/n-1} = S_1^2/S_2^2 \text{ has } F(m-1, n-1)$$

Now plotting λ against F we obtain:



For $\lambda \le \lambda_\alpha$, $F < F_1$ or $F > F_2$

Where F_1 & F_2 satisfy

$$P[F < F_1 \text{ and } F > F_2] = \alpha$$

$$\Rightarrow F_2 = F_{\alpha/2}, \quad F_1 = \frac{1}{F_{\alpha/2}}$$ (reverse order of d.f)

The test is to compute F and reject H_0 whenever $F < F_1$ or $F > F_2$.

Remark:

i) If $H_1: \sigma_1^2 > \sigma_2^2$
compute F and reject H_0 if $F > F_{1-\alpha}$

ii) If $H_1: \sigma_1^2 < \sigma_2^2$
Compute F and reject H_0 if $F < 1/F_\alpha$

iii) $F_{1-\alpha/2}(m,n) = \dfrac{1}{F_{\alpha/2}(n,m)}$

F-test for equal population variances.

One-tailed test	Two tailed test
$H_0: \sigma_1^2 = \sigma_2^2$	$H_0: \sigma_1^2 = \sigma_2^2$
$H_1: \sigma_1^2 < \sigma_2^2$	$H_1: \sigma_1^2 \neq \sigma_2^2$
or ($H_1: \sigma_1^2 > \sigma_2^2$)	
Test statistic:	Test statistic
$F = s_2^2 / s_1^2$	$F = \dfrac{\text{larger sample variance}}{\text{smaller sample variance}}$
or $F = s_1^2 / s_2^2$ when $H_1: \sigma_1^2 > \sigma_2^2$	

	$= s_1^2/s_2^2$ where $s_1^2 > s_2^2$
	or
	s_2^2/s_1^2 where $s_2^2 > s_1^2$
Rejection Region	Rejection Region.
$F > F_{\alpha}(M,n)$	$F > F_{\alpha/2}(M,n)$ or $F < 1/F_{\alpha/2}$.

Assumptions.

1. Both sampled populations are normally distributed.
2. The samples are random & independent

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2.$$

Example:

Independent random samples were selected from each of two normally distributed populations $n_1 = 16$ from population (1) and $n_2 = 25$ from population (2). The means & variances for the two samples are shown in the table below

Sample 1	Sample 2
$n_1 = 16$	$n_2 = 25$
$\bar{x}_1 = 27.5$	$\bar{x}_2 = 28.2$
$s_1^2 = 2.87$	$s_2^2 = 9.85$

a) Test the null hypothesis $H_0: \sigma_1^2 = \sigma_2^2$ against the alternative $H_1: \sigma_1^2 \neq \sigma_2^2$. use $\alpha = 0.05$

b) Test $H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 < \sigma_2^2$ use $\alpha = 0.05$

solution:

a) $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

Two tailed.

Test statistic $F = s_2^2/s_1^2 = 9.85/2.87 = 3.43$

$F_1 = \dfrac{1}{F_{0.025(15,24)}} = 0.41$ ✓

$F_2 = F_{\alpha/2}(24,15) = F_{0.025(24,15)} = 2.70$ ✓

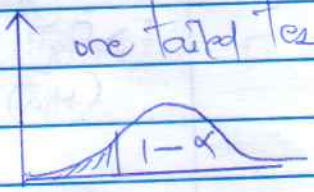
The computed F exceeds the rejection value of 2.70 ie $F > F_{0.025}$

Thus we reject H_0 at 5% level of significance.

b) $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 < \sigma_2^2$

one tailed test



Test statistic $F = s_2^2 / s_1^2 = 3.43$

$$\frac{1}{F_{0.95}(24,15)} = \frac{1}{F_{0.05}(15,24)} = \frac{1}{2.11} = 0.4739$$

Computed F > Ftable value ie $3.43 > 0.4739$

Thus we do not reject H_0 at 5% level of significance.

Testing for Independence in a bivariate Normal population.

Suppose that the r.v.s X & Y have the joint distribution given by bivariate normal population

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right) - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right]}$$

Where $E(X)=\mu$, $Var(X)=\sigma_1^2$, $E(Y)=\mu_2$, $Var(Y)=\sigma_2^2$

$Cov(X,Y)=\rho\sigma_1\sigma_2$

For bivariate normal densities, $\rho=0$, implies that X & Y are independent, a result not true for other distributions. To test the independence of X & Y it is enough to test $\rho=0$, thus we have:

$H_0: \rho=0$ against

$H_1: \rho\neq 0$.

Let $y_1, y_2, \ldots, y_n$ be independent observations on Y such that $y_i = \alpha+\beta x_i + e_i$

The joint distribution $y_1, y_2, \ldots, y_n$ is

$$L(\underline{y}|\underline{x}) = \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\alpha+\beta x_i)^2}$$

We are interested in finding out whether the assumed linear relationship between Y & X exists i.e after finding the best estimators of α & β we must test the hypothesis $H_0: \beta=0$ against $H_1: \beta\neq 0$

The parameter spaces in this case are:

$\Omega = \{(\alpha,\beta,\sigma^2): -\infty<\alpha,\beta<\infty, \sigma^2>0\}$

$\Omega_0 = \{(\alpha,\beta,\sigma^2): -\infty<\alpha<\infty, \beta=0, \sigma^2>0\}$

The MLE's under Ω are: $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(X_i-\bar{X})^2}$$

$$\hat{\sigma}^2 = \sum_{i=1}^{n}(y_i-\hat{\alpha}-\hat{\beta}x_i)^2 / n$$

Under ω_0 the M.L.E's are:

$\hat{\alpha} = \bar{Y}$ since $\beta = 0$, $\hat{\sigma}^2 = \sum_{i=1}^{n}(y_i - \bar{y})^2 / n$

The likelihood ratio is

$$\lambda = \frac{L(\hat{\omega}_0)}{L(\hat{\omega})} = \frac{\left(\frac{1}{2\pi\hat{\sigma}^2}\right)^{n/2} e^{-\frac{1}{2\hat{\sigma}^2}\sum_{i=1}^{n}(y_i - \hat{\alpha})^2}}{\left(\frac{1}{2\pi\hat{\sigma}^2}\right)^{n/2} e^{-\frac{1}{2\hat{\sigma}^2}\sum_{i=1}^{n}(y_i - \hat{\alpha} - \hat{\beta}x_i)^2}}$$

$$\lambda = \left[\frac{\sum_{i=1}^{n}(y_i - \hat{\alpha} - \hat{\beta}x_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}\right]^{n/2}$$

$$\lambda = \left[\frac{\sum_{i=1}^{n}(y_i - \bar{y} + \hat{\beta}\bar{x} - \hat{\beta}x_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}\right]^{n/2}$$

$$= \left[\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2 - 2\hat{\beta}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}^2\sum_{i=1}^{n}(x_i - \bar{x})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}\right]^{n/2}$$

substituting for $\hat{\beta}$ we obtain;

$$\lambda = \left[\sum_{i=1}^{n}(y_i - \bar{y})^2 - 2\frac{\left[\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} + \frac{\left[\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right]^{n/2}$$

$$= \left[1 - \frac{\left[\sum(x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}\right]^{n/2}$$

$$= (1 - r^2)^{n/2}$$

where $r = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$

Since $\hat{\beta}$ is a function of a normal r.v then it is normally distributed with;

$$E(\hat{\beta}) = E\left(\frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})E(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(\cancel{\alpha}+\beta x_i-\beta\bar{x}-\cancel{\alpha})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \beta\sum_{i=1}^{n}(x_i-\bar{x})^2 \Big/ \sum_{i=1}^{n}(x_i-\bar{x})^2 = \beta$$

$$Var(\hat{\beta}) = Var\left[\frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]$$

$$= \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2 Var(Y_i-\bar{Y})}{\left(\sum_{i=1}^{n}(x_i-\bar{x})^2\right)^2} = \sigma^2\sum_{i=1}^{n}(x_i-\bar{x})^2 \Big/ \left(\sum_{i=1}^{n}(x_i-\bar{x})^2\right)^2$$

$$= \sigma^2 \Big/ \sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$\therefore \hat{\beta} \sim N\left(\beta, \sigma^2 \Big/ \sum_{i=1}^{n}(x_i-\bar{x})^2\right)$$

Hence $Z = \dfrac{\hat{\beta}-\beta}{\sqrt{\sigma^2 \Big/ \sum_{i=1}^{n}(x_i-\bar{x})^2}} \sim N(0,1)$, σ^2 is known.

Also $Q = n\hat{\sigma}^2 / \sigma^2 = \sum_{i=1}^{n}(y_i-\hat{\alpha}-\hat{\beta}x_i)^2 / \sigma^2 \sim \chi^2_{(n-2)}$

Hence, $T = \dfrac{Z}{\sqrt{Q/(n-2)}} \sim t_{n-2}$

If H_0 is true, then $\beta = 0$ {

$$T = \frac{\hat{\beta}\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}}{\sqrt{n\hat{\sigma}^2/(n-2)}} \sim t_{(n-2)}$$

Substituting $\hat{\alpha}$, $\hat{\beta}$ and $\hat{\sigma}^2$ in T we have;

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad \text{where } r = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2\sum_{i=1}^{n}(Y_i-\bar{Y})^2}}$$

Hence $\lambda = (1-r^2)^{n/2}$ gives ✓

$$\lambda = \left[\frac{1}{1+T^2/n-2}\right]^{n/2}$$

The test is therefore to compute:

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

and reject H_0 if $|T| > C$

where $P[T>C] = \alpha/2 = P[T<-C]$

Example:

Suppose the following data were observed:

X	2	7	3	5	10	9
Y	1	6	2	5	8	8

Test the hypothesis $H_0: \beta = 0$ against $H_1: \beta \neq 0$ take α to be 5%

Solution:

The test statistic is $T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$

where $r = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2\sum_{i=1}^{n}(Y_i-\bar{Y})^2}}$

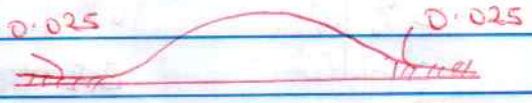
X	Y	$X-\bar{X}$	$Y-\bar{Y}$	$(X-\bar{X})^2$	$(Y-\bar{Y})^2$	$(X-\bar{X})(Y-\bar{Y})$
2	1	-4	-4	16	16	16
7	6	1	1	1	1	1
3	2	-3	-3	9	9	9
5	5	-1	0	1	0	0
10	8	4	3	16	9	12
9	8	3	3	9	9	9
Total 36	30	0	0	52	44	47

$\bar{X} = 30/6 = 6, \quad \bar{Y} = 30/6 = 5$

$$r = 47 \Big/ \sqrt{52 \times 44} = 0.9826$$

$$\therefore T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9826\sqrt{4}}{\sqrt{1-0.9826^2}} = 10.58$$

From table we have; 2.776 $t_{0.025,4}$



Therefore the computed value > table value thus we reject H_0 at 5% level of significance.

The Kruskal-Wallis Test

The K-W test is used to compare Medians among K Comparison groups ($K>2$) and is sometimes described as an ANOVA with data replaced by their ranks.

The H_0 and H_1 for the K-W test are stated as follows:

H_0: The K population Medians are equal vs

H_1: The K population Medians are not equal.

The procedure for the test involves pooling the observations from the K samples into one combined sample, keeping track of which sample each observation comes from, and then ranking lowest to highest from 1 to N, where $N = n_1 + n_2 + \cdots + n_K$.

Example:

A clinical study designed to assess differences in albumin levels in adults following diets with different amounts of protein. Low protein diets are often prescribed for patients with kidney failure. Albumin is the most abundant protein in blood, and its concentration in serum is measured in grams per deciliter (g/dl)

Clinically, serum albumin concentrations are also used to assess whether patients get sufficient protein in their diets. Three diets are compared, varying from 5% to 15% protein, and the 15% protein diet represents a typical American diet. The albumin levels of participants following each diet are shown below

5% protein	10% protein	15% protein
3.1 (4)	3.8 (6)	4.0 (7)
2.6 (1)	4.1 (8)	5.5 (12)
2.9 (2.5)	2.9 (2.5)	5.0 (11)
	3.4 (5)	4.8 (10)
	4.2 (9)	

The sample sizes are small and not equal across comparisons ($n_1=3$, $n_2=5$, $n_3=4$). Thus a non-parametric test is appropriate. The hypotheses to be tested are given below, and we will use a 5% level of significance.

H_0: The three population medians are equal vs
H_1: The three population medians are not equal.

Test statistic for the K-W Test:

The test statistic for the K-W test is denoted H and is defined as follows:

$$H=\left(\frac{12}{N(N+1)}\sum_{j=1}^{K}\frac{R_j^2}{n_j}\right)-3(N+1)$$

Where K = the number of comparison groups, N = the total sample size, n_j is the sample size in the jth group and R_j is the sum of the ranks in the jth group.

$R_1 = 7.5$
$R_2 = 30.5$
$R_3 = 40$

$$H = \frac{12}{12(13)}\left(\frac{7.5^2}{3} + \frac{30.5^2}{5} + \frac{46^2}{4}\right) - 3(13) = 7.52$$

Critical Value of the X^2 Distribution:

The critical value in the table of critical values for the Chi-square distribution for $df = k-1 = 3-1 = 2$ and $\alpha = 0.05$.

The critical value is 5.991

Reject H_0 because $7.52 \geq 5.991$. We have statistically significant evidence at $\alpha = 0.05$ to show that there is a difference in Median protein diets.